

Nested gauges are not lumped rainfall–runoff: a regime model of information value on operational river networks

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Research article submitted to Hydrology and Earth System Sciences

Abstract

Operational nested river gauges are not the same object as lumped rainfall–runoff catchments. Tomorrow’s discharge at a forecast point is already partly written in today’s upstream hydrograph. We estimate when that nested information still has value on 21 documented United States Geological Survey and Water Survey of Canada triads (training through 2018, validation 2019–2020, testing 2021–2023). Instruments are lagged-upstream linear routing, gradient-boosted trees, persistence, an ordinary-least-squares precipitation ceiling, and the critical success index on floods above the training 90th percentile, with block-bootstrap intervals. Extra-gauge skill falls as one-day routing Nash–Sutcliffe efficiency (NSE) approaches one (grouped-climate regime coefficient -0.26, coefficient of determination $R^2 = 0.63$). The James still gains from upstream gauges (change in NSE of +0.111) while the Willamette gains only +0.007; on the Rio Grande the extra-gauge term is negative. Event scores can disagree with NSE. Three-day unused rain is large on some humid main stems and indistinguishable from zero on snowmelt. Nested gauges therefore have a regime of information value; they should not be scored as if they were lumped rainfall–runoff catchments.

Keywords. nested streamflow gauges; information value; flood forecasting; critical success index; precipitation ceiling; operational hydrology

1 Introduction

Catchment rainfall–runoff long short-term memory networks trained on the Catchment Attributes and Meteorology for Large-sample Studies (CAMELS) data set have changed what hydrologists expect from learned models (Kratzert et al., 2018, 2019; Gauch et al., 2021). Extreme-event evaluations have shown that Nash–Sutcliffe efficiency on the bulk hydrograph is not the critical success index on floods (Frame et

al., 2022). Nearing et al. (2021) argued that the interesting question is no longer whether machine learning can beat a lumped conceptual model, but what hydrological science remains once the data are allowed to speak. That literature is about lumped catchments. It is not about the operational nested gauge: a main-stem forecast point that already sees the flood wave in upstream discharge.

Those two objects are routinely scored with the same grammar. Adding a few United States Geological Survey (USGS) example basins and a new encoder does not advance the CAMELS line, and it does not establish when nested information still has value. The missing object is a regime: when lagged-upstream linear routing already saturates next-day skill, when an extra gauge still helps, when unused catchment rain still has a ceiling at 3–7 d, and when flood scores disagree with Nash–Sutcliffe efficiency. The estimate has to be made on many documented networks with uncertainty.

We therefore score 21 nested triads with overlapping daily records and a 2021–2023 test window under a fixed chronological split. Learned instruments stay linear routing, gradient-boosted trees, and persistence. A new architecture is not the claim. CAMELS lumped rainfall–runoff is related work, not a substitute if a nested join fails. Site numbers are given in Table 1.

2 Methods

2.1 Nested networks

Each network is a triad: one target daily mean discharge (USGS parameter 00060, or the Water Survey of Canada equivalent on the Bow) and two upstream gauges on the same basin, joined to Open-Meteo archive precipitation and temperature at the site coordinates. Overlapping records are sought for 1 January 2000 to 31 December 2023. Networks that fail the inner join or that have no 2021–2023 test window are dropped. They are not replaced by CAMELS lumped catchments. Sacramento at Freeport (11447650) was downloaded but ends in September 2015 in this join and is therefore excluded from scoring. Native units are retained (cubic feet per second in the United States; cubic metres per second on the Bow). Climate class, regulation, and ice notes are recorded in Table 1. The split is fixed: training dates through 31 December 2018, validation 2019–2020, test 2021–2023. The random seed is 42.

Network	Climate	Target	Upstream 1	Upstream 2	Units	Regulation / ice
Allegheny	humid Mid-Atlantic	03049500	03036500	03016000	cfs	Kinzua Dam (Allegheny Reservoir); ice: winter ice possible
Animas	snowmelt Rockies	09363500	09358000	09361500	cfs	limited main-stem storage; ice: winter ice possible in headwaters
Bow	snowmelt Rockies (Canada)	05BH004	05BH005	05BB001	m3 s-1	Bearspaw and Ghost reservoirs; ice: WSC Ice Conditions flags in winter
Brazos	subhumid southern Plains	08096500	08089000	08082500	cfs	main-stem reservoirs (Possum Kingdom, Whitney); ice: rare
Chattahoochee	humid Southeast	02338000	02336000	02335000	cfs	Buford Dam / Lake Lanier; ice: none
Deschutes	semiarid interior West	14092500	14091500	14076500	cfs	Pelton-Round Butte; irrigation; ice: rare
Hudson	humid Northeast	01358000	01357500	01335754	cfs	Troy Dam; Mohawk reservoirs; Champlain Canal; ice: winter ice possible
Illinois	humid Midwest	05586100	05568500	05555300	cfs	navigation dams; ice: winter ice possible
Iowa	humid Midwest	05465500	05454500	05453000	cfs	Coralville Dam above Iowa City; ice: winter ice possible
James	humid Mid-Atlantic	02037500	02035000	02034000	cfs	Piedmont reservoirs; ice: rare
Minnesota	humid Midwest	05330000	05325000	05316500	cfs	agricultural drainage; some reservoirs; ice: winter ice common
Potomac	humid Mid-Atlantic	01646500	01636500	01638480	cfs	main-stem reservoirs upstream; urban withdrawals; ice: rare
Rio Grande	semiarid Southwest	08330000	08313000	08289000	cfs	Rio Grande Compact; Cochiti and upstream storage; heavy diversion; ice: rare
Russian	Mediterranean California	11467000	11464000	11461500	cfs	Lake Mendocino / Lake Sonoma; ice: none
Savannah	humid Southeast	02198500	02197000	02192000	cfs	Savannah River dams (Hartwell/Russell/Thurmond system upstream of Augusta); ice: none
Susquehanna	humid Mid-Atlantic	01570500	01536500	01554000	cfs	main-stem dams and reservoirs; ice: occasional
Verde	semiarid Southwest	09506000	09504000	09503700	cfs	diversions; limited storage; ice: none
Wabash	humid Midwest	03377500	03374000	03335500	cfs	limited main-stem storage; agricultural; ice: winter ice possible
Willamette	humid Pacific NW	14191000	14174000	14182500	cfs	Willamette Valley Project reservoirs; ice: rare
Wisconsin	humid Midwest	05407000	05404000	05398000	cfs	paper-mill and hydropower dams; ice: winter ice common
Yakima	semiarid interior West	12510500	12484500	12488500	cfs	Yakima irrigation project reservoirs; ice: occasional

Table 1. Nested triads scored in the atlas (USGS site numbers unless noted; Bow is Water Survey of Canada). Sacramento was downloaded but has no 2021–2023 test window and is omitted.

2.2 Forecast instruments

Persistence uses yesterday's discharge as the forecast issued at the origin. Lagged-upstream linear routing regresses target discharge on contemporaneous local discharge and two upstream hydrographs.

Lag is chosen on the calibration window among $\{0, 1, 2, 3\}$ d. XGBoost (Chen and Guestrin, 2016) uses 300 trees, depth 6, learning rate 0.05, row subsample and column subsample 0.85, and seed 42, trained on 30 d windows of local discharge, precipitation, temperature, and the same upstream series. Ablation removes the upstream columns. The precipitation ceiling is ordinary least squares with perfect future catchment rain as extra features. XGBoost oracle runs are reported in the frozen tables as an overfitting diagnostic when they degrade; they are not used as process evidence. We do not retrain a long short-term memory network on the 21-network atlas; that architecture is a CAMELS rainfall–runoff instrument and is the wrong comparison unless a lumped precipitation-to-discharge control is run as a separate study.

Nash–Sutcliffe efficiency (NSE) on a test series with observations y and predictions $\hat{y}$ is

$$NSE = 1 - \Sigma(y - \hat{y})^2 / \Sigma(y - \bar{y})^2$$

The critical success index (CSI) on floods uses the training-period 90th percentile of target discharge as threshold. With hits H , misses M and false alarms F ,

$$CSI = H / (H + M + F).$$

CSI is interpreted only when the test period has at least 30 events. Iowa (6 events) and Deschutes (19 events) are flagged as low-count. A travel-time proxy is the lag (0–10 d) that maximises Pearson correlation between an upstream hydrograph and the target on the training period. Block bootstrap (14 d blocks, 300 replicates) gives 95 % intervals on ablation Δ NSE, ordinary-least-squares rain Δ NSE, and XGBoost-minus-routing CSI, evaluated on held-out test pairs without refitting.

The regime model is ordinary least squares of 1 d ablation Δ NSE, and separately of 3 d rain-ceiling Δ NSE, on 1 d routing NSE, travel lag, and climate. Twelve climate labels on 21 networks overfit dummy variables; the interpretable specification groups climate as humid, dry/semiarid, snowmelt, or Mediterranean. A previous-run Global Forecast System precipitation swap on 2024 for Potomac, Verde and Animas is reported in the Appendix. It is not used as a headline cap: the ordinary-least-squares oracle is the rain-information test.

Language models were used to draft code and to edit English. Every number in the tables is locked to frozen comma-separated files produced by the scoring scripts.

3 Results

One-day routing NSE on the 21 networks ranges from 0.605 (Verde) to 0.996 (Minnesota). Ablation Δ NSE ranges from -0.016 (Rio Grande) to 0.128 (Russian). Where routing NSE is already near one, extra gauges usually add little next-day skill to a linear hydrograph, and gradient boosting can still lose to routing. The Bow is the clearest example: routing NSE = 0.976 while XGBoost NSE = 0.871, even though upstream columns still help XGBoost relative to a local-only tree. Where routing is weaker, the extra hydrograph can still change next-day skill. That saturation axis is Fig. 1.

Network	Route NSE	XGB NSE	Abl. Δ NSE	Δ NSE 95 % CI	XGB CSI	Route CSI	Lag (d)	Events
Allegheny	0.918	0.963	0.019	[0.010, 0.033]	0.806	0.648	0	64
Animas	0.980	0.984	0.002	[-0.000, 0.006]	0.904	0.914	0	103
Bow	0.976	0.871	0.082	[0.003, 0.234]	0.886	0.855	0	74
Brazos	0.951	0.950	0.011	[0.000, 0.037]	0.808	0.882	7	72
Chattahoochee	0.678	0.779	0.092	[0.051, 0.144]	0.546	0.445	1	146
Deschutes*	0.880	0.875	0.010	[-0.001, 0.023]	0.636	0.522	1	19
Hudson	0.788	0.871	0.013	[0.004, 0.020]	0.644	0.523	0	106
Illinois	0.991	0.994	0.001	[0.001, 0.002]	0.900	0.872	2	46
Iowa*	0.966	0.973	0.014	[0.006, 0.023]	0.333	0.250	0	6
James	0.920	0.941	0.111	[0.068, 0.180]	0.845	0.775	0	77
Minnesota	0.996	0.997	0.006	[0.001, 0.012]	0.959	0.933	2	72
Potomac	0.861	0.920	0.051	[0.015, 0.123]	0.766	0.622	0	39
Rio Grande	0.987	0.968	-0.016	[-0.043, -0.002]	0.832	0.817	1	135
Russian	0.939	0.898	0.128	[0.015, 0.367]	0.805	0.861	0	68
Savannah	0.985	0.995	0.011	[0.007, 0.017]	0.936	0.866	7	154
Susquehanna	0.842	0.949	0.061	[0.024, 0.120]	0.802	0.615	0	72
Verde	0.605	0.678	0.113	[-0.095, 0.347]	0.828	0.540	0	139
Wabash	0.987	0.993	0.003	[0.002, 0.005]	0.924	0.897	1	75
Willamette	0.970	0.985	0.007	[0.001, 0.013]	0.835	0.807	0	96
Wisconsin	0.986	0.991	0.028	[0.020, 0.039]	0.899	0.850	2	71
Yakima	0.953	0.983	0.020	[0.008, 0.043]	0.808	0.867	2	70

Table 2. One-day nested skill on the 2021–2023 test window. Ablation is full XGBoost minus local-only. Intervals are 14 d block bootstrap. Asterisks mark fewer than 30 P90 events. Units of discharge follow Table 1.

The James versus Willamette contrast survives the larger atlas. James ablation $\Delta\text{NSE} = 0.111$ [0.068, 0.180]; Willamette $\Delta\text{NSE} = 0.007$ [0.001, 0.013]. Both intervals exclude zero, but the James extra-gauge term is more than ten times larger. The Rio Grande extra-gauge term is negative (-0.016 [-0.043, -0.002]): persistence and routing already explain tomorrow, and the extra columns do not help the tree model. On the Verde, NSE and CSI disagree in size: XGBoost NSE = 0.678 versus routing 0.605, while XGBoost CSI = 0.828 versus routing 0.540. Event-oriented evaluation scored only on NSE would understate the flood-event gain.

The grouped-climate regime for 1 d ablation ($n = 21$, $R^2 = 0.627$) has routing-NSE coefficient -0.261. Eleven of 21 networks have a travel-time proxy of 0 d; daily correlation with the target is already high, so lag is unresolved at this time step. Mediterranean California is the climate group with the largest residual extra-gauge term after saturation is accounted for.

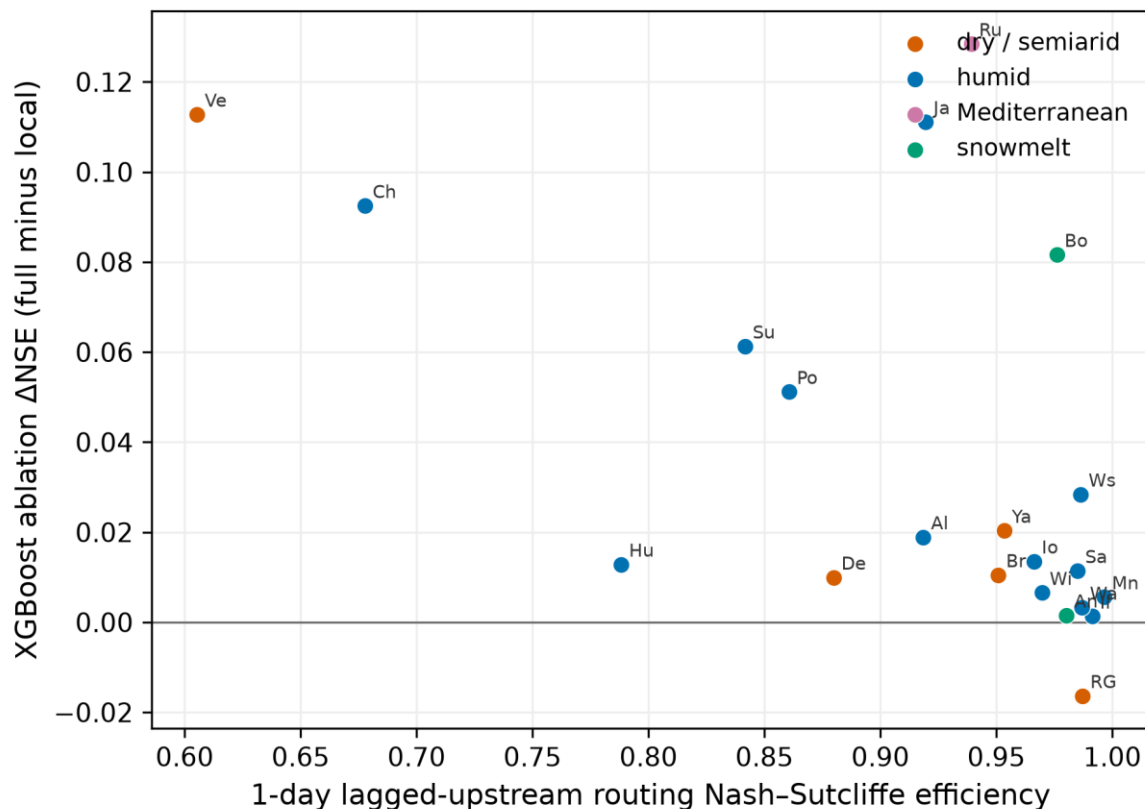


Figure 1. One-day XGBoost ablation ΔNSE against lagged-upstream routing NSE. Colours are climate groups (Okabe–Ito). Labels are two-letter codes of the Table 1 names (Al Allegheny, An Animas, Bo Bow, Br Brazos, Ch Chattahoochee, De Deschutes, Hu Hudson, Il Illinois, Io Iowa, Ja James, Mn Minnesota, Po Potomac, RG Rio

Grande, Ru Russian, Sa Savannah, Su Susquehanna, Ve Verde, Wa Wabash, Wi Willamette, Ws Wisconsin, Ya Yakima). Ablation Δ NSE tends to fall as routing NSE rises; Bow remains an exception on the vertical axis because upstream columns still help the tree relative to a local-only model.

The 3 d ordinary-least-squares precipitation ceiling is the instrument-independent rain test (Table 3, Fig. 2). Hudson Δ NSE = 0.222 [0.120, 0.316]. Animas Δ NSE = 0.002 [-0.001, 0.008]; Bow Δ NSE = -0.009 [-0.040, 0.009]. Unused 3 d rain is a humid-main-stem property in this sample, not a universal mid-range deficit. The grouped rain-ceiling regime has routing-NSE coefficient -0.386 ($R^2 = 0.666$): once 1 d routing is saturated, perfect future rain also has little left to give at 3 d.

Network	OLS NSE	Oracle NSE	Δ NSE	Δ NSE 95 % CI
Allegheny	0.738	0.793	0.055	[0.013, 0.086]
Animas	0.921	0.923	0.002	[-0.001, 0.008]
Bow	0.895	0.886	-0.009	[-0.040, 0.009]
Brazos	0.875	0.845	-0.029	[-0.314, -0.009]
Chattahoochee	0.263	0.402	0.139	[-0.055, 0.280]
Deschutes	0.489	0.523	0.034	[-0.016, 0.112]
Hudson	0.413	0.635	0.222	[0.120, 0.316]
Illinois	0.958	0.965	0.007	[0.002, 0.014]
Iowa	0.832	0.816	-0.016	[-0.058, 0.013]
James	0.539	0.596	0.057	[-0.033, 0.143]
Minnesota	0.986	0.985	-0.001	[-0.006, 0.002]
Potomac	0.333	0.407	0.074	[-0.118, 0.160]
Rio Grande	0.947	0.943	-0.003	[-0.007, 0.001]
Russian	0.529	0.628	0.099	[-0.414, 0.280]
Savannah	0.954	0.956	0.002	[0.001, 0.004]
Susquehanna	0.531	0.696	0.166	[0.067, 0.271]
Verde	-0.064	0.011	0.075	[0.048, 0.530]
Wabash	0.935	0.955	0.020	[0.011, 0.032]
Willamette	0.790	0.888	0.099	[0.052, 0.137]
Wisconsin	0.965	0.968	0.003	[-0.002, 0.011]
Yakima	0.890	0.895	0.005	[-0.004, 0.016]

Table 3. Three-day ordinary-least-squares precipitation ceiling on the 2021–2023 test window.

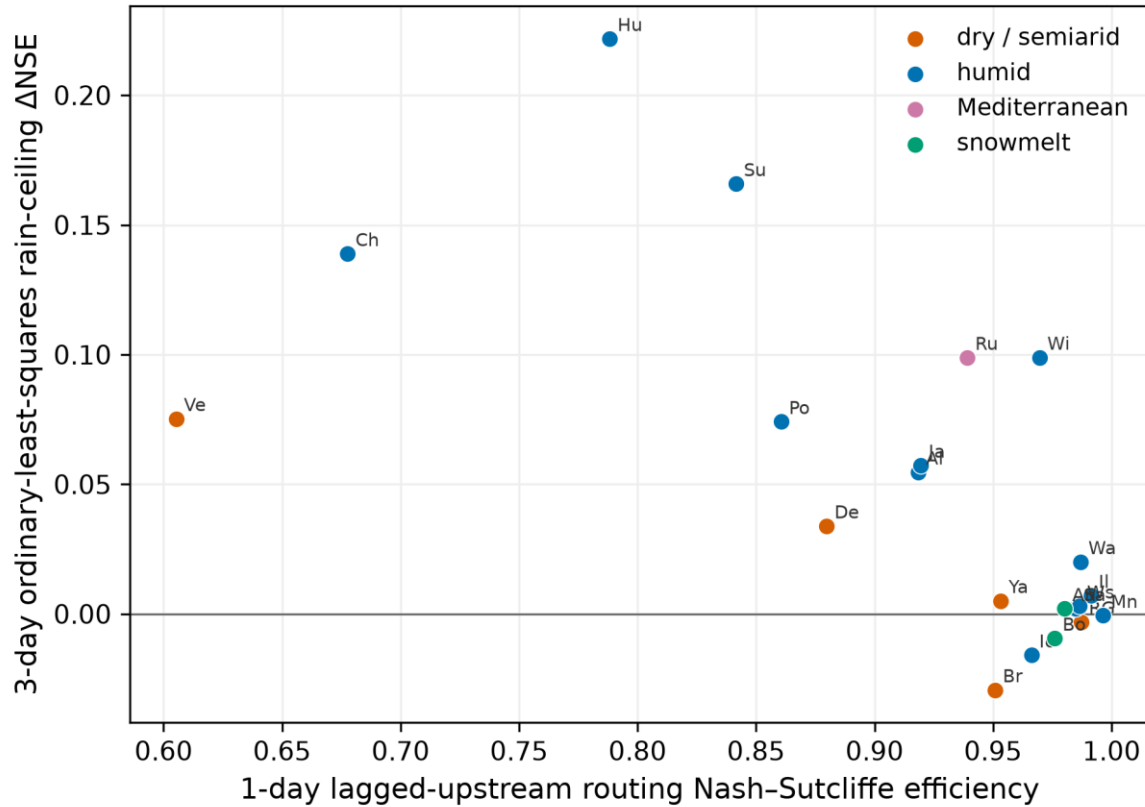


Figure 2. Three-day ordinary-least-squares rain-ceiling Δ NSE against 1-day routing NSE. Labels follow Fig. 1. Snowmelt and already-saturated main stems sit near zero; some humid stems still have unused 3 d rain.

4 Discussion

Kratzert et al. (2018) showed that a regional long short-term memory network can beat lumped conceptual models on CAMELS. Frame et al. (2022) showed that NSE on the bulk hydrograph is not CSI on extremes. Nearing et al. (2021) asked what science remains after the machine-learning win. None of those papers is a nested-gauge information-value atlas. Mixing CAMELS lumped basins into this sample would erase the distinction the paper exists to make.

A practical reading follows from the regime. If 1 d routing NSE is already high, another encoder is the wrong investment; the hydrograph is already in the upstream gauge, and a tree model can be worse than routing (Bow). If ablation Δ NSE remains clearly positive, the extra nested record still adds skill (James, Russian). If the extra-gauge term is negative, more columns are not more information (Rio Grande). If the ordinary-least-squares rain ceiling is near zero, 3–7 d precipitation cannot rescue the

forecast (Animas, Bow, Minnesota). If CSI and NSE disagree, event-oriented evaluation should not rely on NSE alone (Verde). Those statements require a sample of networks and uncertainty intervals, not a few case studies.

Limits: regulation and ice are noted, not modelled as processes; travel time is unresolved at daily resolution on most of these main stems; Iowa and Deschutes CSI rest on too few events; a 2024 Global Forecast System previous-run swap on three climates (Appendix) does not reproduce a systematic operational precipitation penalty and is therefore not a headline cap. NeuralHydrology entity-aware long short-term memory remains a rainfall–runoff comparison for a different paper.

5 Conclusions

Nested operational gauges are not lumped rainfall–runoff. Information value on those gauges is a regime in routing saturation, climate group, and event versus bulk scores, estimated here with block-bootstrap uncertainty on 21 documented triads. The contribution is that regime.

Appendix A: Additional leads and 2024 precipitation swap

Tables A1 and A2 repeat the XGBoost versus routing comparison at 3 d and 7 d. Skill falls with lead on almost every network; CSI falls faster than NSE on several humid stems (Potomac, Susquehanna, Hudson). Table A3 reports a 2024 test in which the local precipitation feature is swapped for Open-Meteo previous-run Global Forecast System daily precipitation after training on observed rain. The swap does not systematically undercut observed-rain skill. On the Verde the previous-run field even raises 3 d and 7 d NSE relative to the Open-Meteo archive observation. That is not evidence that operational precipitation is sufficient; it is evidence that this particular swap is not the information bottleneck on those 2024 samples. The ordinary-least-squares oracle in Table 3 remains the rain-information test.

Network	Route NSE	XGB NSE	Abl. ΔNSE	Route CSI	XGB CSI	Events
Allegheny	0.678	0.752	0.019	0.225	0.325	64
Animas	0.904	0.912	-0.000	0.886	0.829	103
Bow	0.896	0.916	0.210	0.795	0.783	74
Brazos	0.895	0.871	0.025	0.868	0.720	72
Chattahoochee	0.187	0.190	0.010	0.105	0.216	146
Deschutes	0.556	0.432	0.046	0.269	0.292	19
Hudson	0.399	0.444	0.007	0.152	0.245	106
Illinois	0.945	0.957	0.011	0.636	0.678	46
Iowa	0.839	0.833	0.066	0.000	0.111	6

James	0.435	0.479	0.016	0.256	0.364	77
Minnesota	0.971	0.976	0.042	0.846	0.883	72
Potomac	0.380	0.465	0.060	0.200	0.328	39
Rio Grande	0.947	0.921	-0.019	0.622	0.655	135
Russian	0.647	0.517	0.042	0.537	0.439	68
Savannah	0.910	0.972	0.116	0.689	0.853	154
Susquehanna	0.442	0.485	0.014	0.198	0.315	72
Verde	0.245	0.242	0.160	0.553	0.513	139
Wabash	0.916	0.939	0.032	0.695	0.792	75
Willamette	0.771	0.805	0.010	0.520	0.610	96
Wisconsin	0.939	0.954	0.132	0.653	0.764	71
Yakima	0.768	0.873	0.057	0.619	0.667	70

Table A1. 3-day nested skill on the 2021–2023 test window.

Network	Route NSE	XGB NSE	Abl. Δ NSE	Route CSI	XGB CSI	Events
Allegheny	0.385	0.499	0.034	0.070	0.127	64
Animas	0.848	0.847	-0.005	0.687	0.687	103
Bow	0.729	0.857	0.037	0.505	0.618	74
Brazos	0.791	0.688	0.019	0.675	0.737	72
Chattahoochee	0.130	0.150	0.004	0.086	0.127	146
Deschutes	0.280	0.264	0.298	0.042	0.120	19
Hudson	0.292	0.315	0.027	0.024	0.079	106
Illinois	0.811	0.814	0.050	0.509	0.514	46
Iowa	0.606	0.599	0.094	0.000	0.000	6
James	0.235	0.288	-0.002	0.092	0.160	77
Minnesota	0.790	0.819	0.094	0.675	0.651	72
Potomac	0.072	0.110	0.087	0.038	0.075	39
Rio Grande	0.881	0.856	0.025	0.519	0.576	135
Russian	0.328	0.358	0.118	0.337	0.383	68
Savannah	0.814	0.850	0.399	0.758	0.800	154
Susquehanna	0.175	0.160	-0.004	0.012	0.078	72
Verde	0.188	0.198	0.070	0.494	0.412	139
Wabash	0.686	0.742	0.058	0.378	0.409	75
Willamette	0.510	0.511	-0.016	0.290	0.317	96
Wisconsin	0.658	0.653	0.054	0.409	0.422	71
Yakima	0.543	0.567	-0.003	0.211	0.384	70

Table A2. 7-day nested skill on the 2021–2023 test window.

Network	Lead (d)	Observed-rain NSE	Previous-run NSE	Δ	n (2024)
Animas	1	0.961	0.961	-0.000	348
Animas	3	0.830	0.828	-0.002	346
Animas	7	0.611	0.615	0.004	342
Potomac	1	0.939	0.937	-0.002	348
Potomac	3	0.482	0.515	0.033	346
Potomac	7	0.174	0.204	0.030	342
Verde	1	0.694	0.736	0.042	348
Verde	3	0.247	0.410	0.163	346
Verde	7	0.001	0.305	0.304	342

Table A3. Previous-run Global Forecast System precipitation swap on 2024. Models are trained on observed precipitation; only the 2024 local rain feature is replaced.

Code and data availability

Frozen atlas tables and the Word builder are in the forecast-information-value repository (<https://github.com/Az0998/forecast-information-value>, directory data/frozen/river/). Daily discharge is from USGS National Water Information System parameter 00060 and from the Water Survey of Canada for the Bow. Meteorology and previous-run precipitation are from Open-Meteo. Scoring scripts live in the

sibling engine hydro-ml-paper (download_nested_atlas.py, run_nested_atlas.py). Training dates run through 31 December 2018. A persistent identifier for a versioned snapshot will be issued on Zenodo at submission; until then the GitHub frozen tables are the lock file.

Author contributions

SZ designed the study, performed the analysis, and wrote the manuscript.

Competing interests

The author declares that they have no conflict of interest.

Acknowledgements

Discharge records were provided by the USGS National Water Information System and the Water Survey of Canada. Meteorological fields were provided by Open-Meteo. Language models were used to draft analysis code and to edit English; the author remains responsible for the text and for every number locked to the frozen tables.

Financial support

This research received no external financial support.

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